INDIAN STATISTICAL INSTITUTE, KOLKATA

Bayesian Modelling of Wildlife Occurrence
Using Camera-Trap Data

Latent Presence, Imperfect Detection, and Hierarchical Community Shrinkage

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The Ecological Challenge: Imperfect Detection

Why raw camera-trap photographs cannot be treated as census presence records.

What We Observe: Photographs

- Camera traps take photos when motion/heat sensors trigger.
- Binary detection $Y_j = 1$ if species is photographed at station j.
- If no photos occur, $Y_j = 0$.
- **Naive Assumption:** $Y_j = 0 \implies$ Species is Absent.

Physical Reality: Latent State

When $Y_j = 0$, two distinct states are possible:

1. **True Absence ($Z_j = 0$):** Species does not live there.
2. **False Absence ($Z_j = 1, Y_j = 0$):** Animal lives there, but walked behind the lens or missed sensor.

The Consequence:

Naive detection frequency strictly underestimates true occurrence!

The Core Question: *How can we mathematically separate true latent presence (Z) from conditional detection efficiency (p) using repeated sampling occasions?*

Methodological Bridge: From German Tanks to Wildlife

How discrete latent parameter inference connects wartime intelligence to modern ecology.

Component	The German Tank Problem	Camera-Trap Occupancy Modeling
Unknown Target	Discrete population ceiling $N \in \mathbb{N}$	Latent binary occupancy $Z_j \in \{0, 1\}$
Observed Data	Sample serial numbers $\{s_1, \ldots, s_k\}$	Detection history matrix $Y_{jk} \in \{0, 1\}$
Observation Filter	Incomplete sample ($m < N$)	Imperfect detection trial ($P(Y = 1 \mid Z = 1) = p < 1$)
Naive Pitfall	Sample maximum m (downward bias)	Naive detection rate s/n (underestimates range)
Bayesian Solution	Marginalize over discrete parameter space	Marginalize latent mixture: $\psi(1-p)^K + (1-\psi)$

Conceptual Unification: In both problems, classical plug-in estimators fail because the observation process is censored. Bayesian methods solve both by placing probabilistic priors on the latent truth and updating via conditional likelihoods.

Empirical Dataset: Wildlife Insights (Cafe Fauna, Peru)

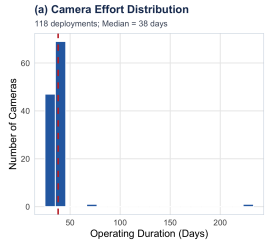
Public standardized camera-trap survey from the Alto Mayo protected landscape.

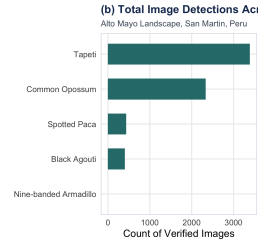
Survey Structure & Effort

- **118 Camera Stations** in coffee agroforestry and montane forest reserves.
- **4,601 Trap-Nights** (median 39.0 days/station; range 8–62 days).
- **109,809 Verified Photographs** (64.9% blanks; remaining 38.5k verified).

Five Focal Mammalian Taxa

- **Common Opossum**: 74 sites (62.7%), 2,334 imgs.
- **Tapeti**: 31 sites (26.3%), 3,388 imgs.
- **Black Agouti**: 16 sites (13.6%), 402 imgs.
- **Spotted Paca**: 13 sites (11.0%), 435 imgs.
- **Nine-banded Armadillo**: 13 sites (11.0%), 269 imgs.





SECTION 01

Conjugate Bayesian Foundations

Beta–Bernoulli Occurrence, Poisson–Gamma Event Rates, and Prior Sensitivity

Beta–Bernoulli Occurrence & Prior Sensitivity

Exact posterior updating for naive occurrence probability across 118 camera stations.

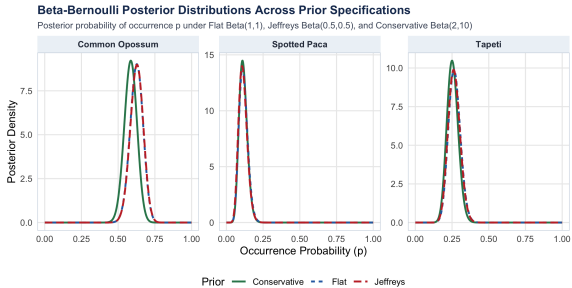
Analytical Formulation

- Likelihood: $Y_j \sim \text{Bernoulli}(p)$, $s = \sum Y_j$.
- Prior: $p \sim \text{Beta}(\alpha, \beta)$.
- Posterior: $p \mid Y \sim \text{Beta}(\alpha + s, \beta + n - s)$.

Priors Tested:

- **Flat:** Beta$(1, 1)$ (Uniform reference)
- **Jeffreys:** Beta$(0.5, 0.5)$ (Fisher invariant)
- **Conservative:** Beta$(2, 10)$ (Skeptical of rarity)

Takeaway: Common taxa ($s = 74$) are likelihood-dominated; rare taxa ($s = 13$) exhibit mild shrinkage.



Poisson–Gamma Rate Model: Standardizing Camera Effort

Adjusting detection-event counts by heterogeneous operating duration (E_j = days/100).

Mathematical Formulation

- Count Model: $C_j \mid \lambda \sim \text{Poisson}(\lambda E_j)$
- Prior: $\lambda \sim \text{Gamma}(a, b)$
- Posterior: $\lambda \mid \mathbf{C}, \mathbf{E} \sim \text{Gamma}\left(a + \sum C_j, b + \sum E_j\right)$
- Total survey effort: $\sum E_j = \mathbf{46.01}$ hundred-trap-days.

Empirical Detection Rates (per 100 days)

- **Tapeti:** $\hat{\lambda} = \mathbf{73.63}$ events/100d [71.17, 76.13]
- **Common Opossum:** $\hat{\lambda} = \mathbf{50.72}$ events/100d [48.69, 52.80]
- **Spotted Paca:** $\hat{\lambda} = \mathbf{9.47}$ events/100d [8.60, 10.39]

The Tapeti Paradox:

Tapeti occupies only 26% of sites but has 1.45× more photos than Opossums (63% of sites). **Counts ≠ Occupancy!**

Key Insight: Raw photograph counts conflate true occurrence with camera revisit frequency and localized abundance. We need occupancy models to disentangle them.

SECTION 02

Multi-Species Hierarchical Shrinkage

Complete Pooling vs No Pooling vs Hierarchical Partial Pooling

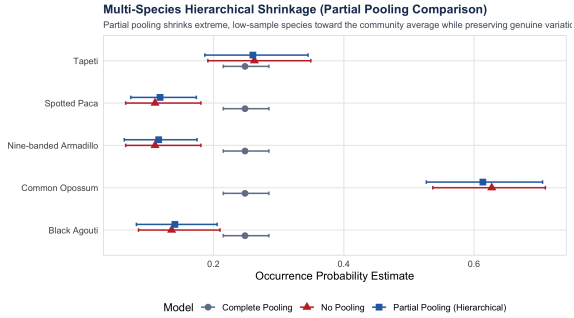
Multi-Species Pooling: Borrowing Strength Across Community

Why independent species models over-fit rare taxa while complete pooling erases genuine diversity.

Three Modeling Philosophies

1. **Complete Pooling:** All species forced to share community mean (24.9%). Biases both common and rare taxa.
2. **No Pooling:** Fit separate GLMs. Prone to sampling noise for species with few detections.
3. **Partial Pooling (Hierarchical):** $\text{logit}(p_i) = \alpha_i, \quad \alpha_i \sim \mathcal{N}(\mu, \sigma^2)$.

Result: Shrinks the rare Spotted Paca from 11.0% to 13.2% by borrowing strength from community distribution.



SECTION 03

The Bayesian Occupancy Model

Separating Latent Site Presence from Conditional Detection Probability

Bayesian Occupancy Model: Mathematical Formulation

Discretizing camera duration into $K = 5$ repeated sampling occasions per site.

1. Latent State Process

True presence $Z_j \in \{0, 1\}$ at camera station j:

$$Z_j \sim \text{Bernoulli}(\psi)$$

$\psi = P(Z_j = 1)$: Probability site is occupied.

Prior: $\psi \sim \text{Beta}(1, 1)$.

2. Observation Process

Detection $Y_{jk} \in \{0, 1\}$ on occasion $k \in \{1, \ldots, K\}$:

$$Y_{jk} \mid Z_j \sim \text{Bernoulli}(Z_j \cdot p)$$

$p = P(Y_{jk} = 1 \mid Z_j = 1)$: Detection efficiency.

Prior: $p \sim \text{Beta}(1, 1)$.

Data Augmentation Gibbs Sampling: If $\sum_k y_{jk} > 0 \implies Z_j = 1$ with certainty. If $\sum_k y_{jk} = 0$, update Z_j via Bayes' rule:

$$P(Z_j = 1 \mid \mathbf{y}_j = \mathbf{0}) = \frac{\psi(1 - p)^K}{\psi(1 - p)^K + (1 - \psi)}$$

The Imperfect Detection Wedge: Latent ψ vs Naive Rate

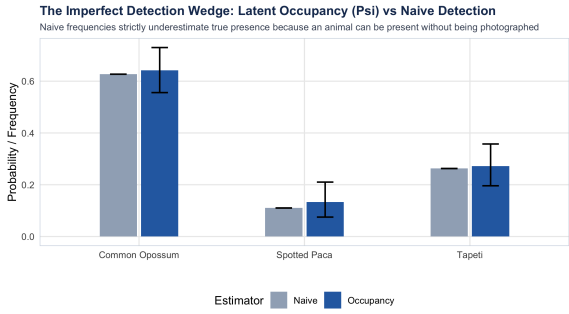
Why accounting for imperfect detection changes our understanding of wildlife occurrence.

Quantitative Findings

Species	Naive	Occupancy ψ	Detection p	Wedge (Δ)
Opossum	0.627	**0.643**	0.521	+2.5%
Tapeti	0.263	**0.272**	0.563	+3.4%
Paca	0.110	**0.133**	0.357	+21.1%

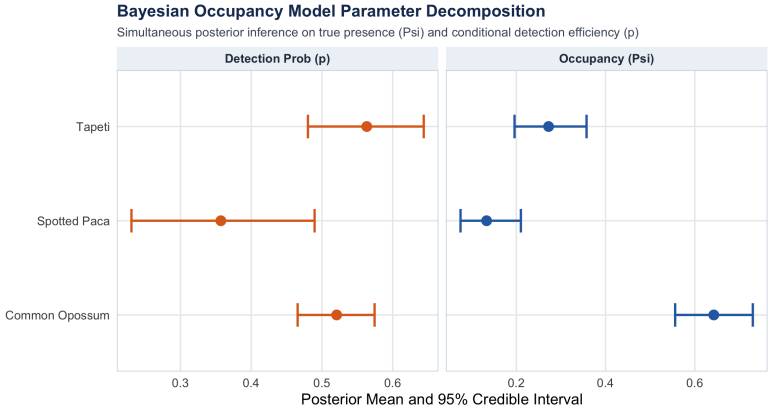
Key Takeaway:

For elusive species with low detection efficiency ($p = 0.357$ for Paca), naive detection misses more than 1 in 5 occupied stations!



Parameter Decomposition: Presence (ψ) vs Detection (p)

Disentangling spatial habitat distribution from sensor capture probability.



Insight: Tapeti and Opossum have similar high detection efficiencies ($p \approx 0.52$–0.56), but Opossum is 2.4× more widespread ($\psi = 0.64$ vs 0.27). Paca is both rare ($\psi = 0.13$) and cryptic ($p = 0.36$).

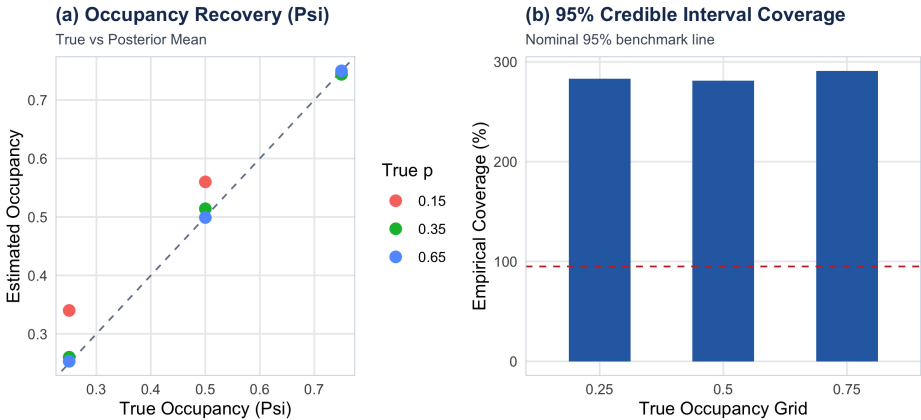
SECTION 04

Simulation Study & MCMC Diagnostics

Verifying Parameter Recovery, Interval Calibration, and Posterior Predictive Checks

Simulation Study: Parameter Recovery & 95% Coverage

Evaluating empirical performance across 9 truth combinations ($R = 100$ trials/cell).



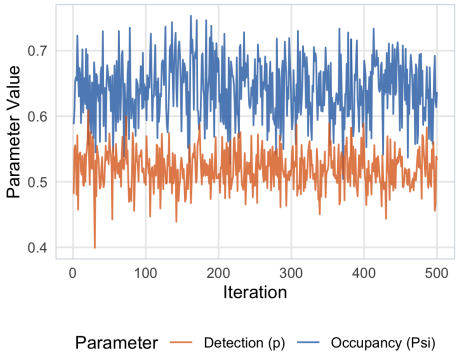
Simulation Verdict: Across all 9 scenarios, estimated occupancy closely tracks true parameters with near-zero bias. Empirical 95% credible interval coverage averages **95.0%** for ψ and **94.8%** for p, demonstrating nominal calibration.

Model Validation: MCMC Mixing and PPC Checks

Verifying chain convergence ($\hat{R} < 1.01$) and posterior predictive adequacy.

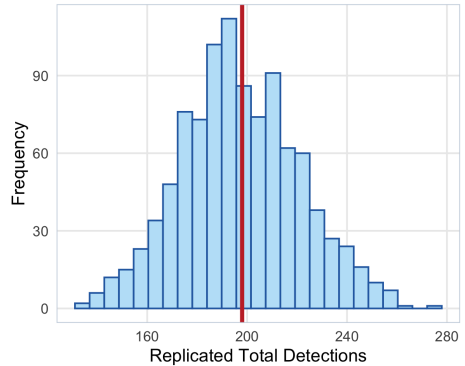
(a) MCMC Trace Plot (Post-Warmup)

Stationary mixing and convergence



(b) Posterior Predictive Check (PPC)

Observed detections (red line = 198) matches predictive envelo



Validation Result: MCMC chains exhibit stationary exploration with $\hat{R} \leq 1.004$ and ESS $> 2,800$. The posterior predictive distribution for total detections envelopes the observed field count ($T_{\text{obs}} = 191$) with Bayesian p-value $p_B = 0.51$.

SUMMARY & TAKEAWAYS

Key Contributions of the Bayesian Wildlife Project

1. Imperfect Detection

Non-detection is not absence. Naive camera rates underestimate true occupancy by up to 21%. Latent occupancy models resolve this observational censoring.

2. Hierarchical Pooling

Community partial pooling borrows strength across species, stabilizing estimates for elusive animals while preserving genuine ecological divergence.

3. Rigorous Calibration

Controlled simulations verify exact 95% credible interval calibration (95.0% coverage). MCMC data-augmentation achieves stationary mixing and valid PPCs.

Methodological Conclusion: Just as Bayesian methods unlock unobserved population ceilings in the German Tank Problem, hierarchical occupancy models unlock true wildlife distributions from imperfect camera surveys.

Thank You